MIL-H-36266A(DSA-DM)
11 May 1966
SUPERSEDING
MIL-H-36266(DM)
18 August 1964

MILITARY SPECIFICATION

HYDROMETERS, GRADUATED SCALE
A.P.I. GRAVITY

1. SCOPE

1.1 Scope. This specification covers glass graduated scale hydrometers suitable for determining the specific gravity of crude petroleum or petroleum products normally handled as liquids and having a surface tension not exceeding 33 dynes per centimeter.

1.2 Classification. Shall be American Petroleum Institute (A.P.I.) hydrometers of the following American Society For Testing And Materials (ASTM) hydrometer numbers, as specified (see 6.1):

Class A. Length 330 millimeters (mm); diameter 25 mm.

ASTM NO:	
1H-62	6H-62
2H-62	7H-62
3H-62	8H-62
4H-62	9H-62
5H-62	10H-62

Class B. Length 163 (mm); diameter 13.5 mm.

ASTM NO:	
21H-62	31H-62
22H-62	32H-62
23H-62	33H-62
24H-62	34H-62
25H-62	35H-62
26H-62	36H-62
27H-62	37H-62
28H-62	38H-62
29H-62	39H-62
30H-62	40H-62

FSC 6630

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials

ASTM Designation E100-62 ASTM Hydrometers

ASTM Designation E126-62 Inspection, Test, and Standardization of Hydrometers

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials. The materials used in the construction of hydrometers covered by this specification shall be in accordance with ASTM Designation E100-62.

3.2 Construction. Class-A and Class-B hydrometers shall be constructed in accordance with ASTM Designation E100-62.

3.3 Dimensions.

3.3.1 Class A hydrometers. The overall length of Class A hydrometers shall be 330 ± 5 mm, and diameter of the body shall be 25 ± 2 mm. All other dimensions shall be as specified in ASTM Designation E100-62 for the appropriate ASTM Hydrometer.

3.3.2 Class B hydrometers. The overall length of Class B hydrometers shall be 163 ± 5 mm and the diameter of the body shall be 13.5 ± 1.5 mm. All other dimensions shall be as specified in ASTM Designation E100-62 for the appropriate ASTM Hydrometer.

3.4 Accuracy.

3.4.1 Class-A. The error in the A.P.I. gravity scale shall not be greater than 0.1° A.P.I., when tested as specified in 4.4.

3.4.2 Class-B. The error in the A.P.I. gravity scale shall not be greater than 0.2° A.P.I., when tested as specified in 4.4.

3.5 Identification marking. The following information shall be legibly and permanently marked in a readily visible location on the inside of the hydrometer: units of hydrometer scale, standard temperature (60° F.), ASTM Hydrometer No., manufacturer's serial number, and the manufacturer's name or registered trademark. Markings secured by adhesives shall not be permitted.

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3.6 Workmanship. Workmanship shall be first class throughout. The hydrometer shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	4.0

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4.2.2 For test. Sampling for accuracy test shall be conducted in accordance with MIL-STD-105 for inspection level S-1. Acceptance number shall be zero.

4.3 Examination. The hydrometers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major		
	101	Ballast not secure in hydrometer.
	102	Scale not legible, not correct, or not securely fixed in hydrometer.
	103	Hydrometer not dry inside.
	104	Glass not transparent, not free of striae, or not smooth.
	105	Stem and body not symmetrical to the vertical axis.
Minor		
	201	Marking not correct, legible, or permanent.
	202	Hydrometers not free from dirt, grease or other foreign material.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The hydrometers shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. The hydrometers shall be tested to determine the accuracy in accordance with ASTM Designation E126-62.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One hydrometer, of the ASTM number specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	48 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. In addition, the ASTM number, name and range shall be shown on the end of the carton.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with Military Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior containers. Exterior containers shall be marked in accordance with Military Standard MIL-STD-129. Marking shall include the legend: "GLASS - HANDLE WITH CARE."

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) ASTM Hydrometer No. and Class required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, or sizes, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types normally procured to meet Military requirements.

Custodians:
Army. MU
Navy. SH
Air Force. 67

Preparing activity:
Defense Supply Agency. DM

Review activities:
Army. MU
Navy. SH
Air Force. 67

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-36266A(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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